

Shruti Jayaraman

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EDUCATION

University of Michigan

Ann Arbor, MI

Master of Science, Computer Science and Engineering | GPA: 3.7/4.0

Aug 2025 – Expected May 2027

Coursework: NLP, Randomness and Computation, Scalable Systems for Agentic AI, Computer Vision, Open Source Innovation & Impact

College of Engineering, Guindy (Anna University)

Chennai, India

Bachelor of Engineering, Computer Science and Engineering | GPA: 9.3/10

Jul 2019 – Apr 2023

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, HTML/CSS, SQL, Bash

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, LLMs, RAG, LangGraph, Stable Diffusion, scikit-learn, pandas, NumPy

Tools & Platforms: Git, FastAPI, LangChain, Streamlit, ROS2, Nav2, Gazebo, Boto3 (AWS Python SDK), Linux

Databases: MySQL, Oracle, ChromaDB, SQLite

PROFESSIONAL EXPERIENCE

Caterpillar Inc.

Chennai, India

Automation Engineer

Jan 2023 – May 2025

- Validated a **GNSS/IMU sensor fusion system** for autonomous mining vehicles operating in **high-altitude and underground environments**, achieving **15% latency reduction** in real-time navigation and improving positional accuracy in GPS-degraded conditions
- Integrated **Bullet3 and ODE physics engines** into a proprietary simulation platform to model complex underground terrain dynamics, enabling validation of autonomous vehicle behavior in simulated environments and reducing dependence on physical test site deployments
- Engineered a **geospatial ETL pipeline** in Python producing interactive map visualizations for mine site planning, and an **automated validation system** flagging discrepancies across **4 databases (12,000+ records each)**, cutting manual validation effort by **60%**

Lavu Inc.

Chennai, India

DevOps Analyst Intern

Jun 2022 – Aug 2022

- Implemented an **anomaly detection pipeline** using **K-Means clustering** on **10,000+ AWS CloudWatch logs**, enabling real-time identification of irregular traffic patterns and eliminating manual log monitoring overhead

RESEARCH EXPERIENCE

Blablablab, University of Michigan

Ann Arbor, MI

Student Researcher (advised by Prof. David Jurgens)

Jun 2026 – Present

- Extending an **intersectional LLM bias study** toward publication; designing a **blind self-evaluation paradigm** testing whether models judge their own outputs more leniently, authored an **11-category bias codebook** with intersectional coding guidance

LIT Lab, University of Michigan

Ann Arbor, MI

Student Researcher (advised by Prof. Rada Mihalcea)

Oct 2025 – Present

- Leading an independent study on which components of **agent skill files** (Claude Code, OpenClaw) drive **agent execution fidelity**; formulated a **span-level attribution** methodology evaluated on **deterministic agentic tasks** with open-source models
- Contributed to an **ACL-submitted clinical NLP system** for medical visit preparation, co-designing a **RAG-based evaluation framework** across **20 human and 90 simulated patients**; reduced patient mental effort (**4.80 → 3.90, p = 0.014**) vs. a plain LLM baseline

PUBLICATIONS

TexAVI: Generating Stereoscopic VR Video Clips From Text Descriptions

Jul 2023

- First-authored a **novel text-to-stereoscopic VR video pipeline** (published at **IEEE CVMI 2024**) integrating **AttnGAN**, **Stable Diffusion**, and **MiDaS depth estimation** to generate immersive VR clips from text prompts

PROJECTS

MemHub: A Memory-as-a-Service Solution for Agentic AI

Apr 2026

- Introduced a **Memory-as-a-Service layer** for multi-agent LLM systems to solve context-window overflow and cross-agent knowledge silos
- Constructed a **two-tier SQLite + ChromaDB architecture** with **confidence-gated retrieval**, achieving **6% lower latency**, **25% tighter memory budget adherence**, and **2.5× more accessible historical context** vs. a single-tier baseline across **7 benchmarks**

RepoRanger: Privacy-first AI Code Reviewer & Automated Branch Janitor

Apr 2026

- Launched a **Dispatcher-Worker GitHub App** with **AI-powered PR reviews** (Groq/Llama-3.3-70b) and keyword-triggered **automated branch hygiene** running entirely on user infrastructure via **GitHub Actions** to reduce dependency on paid code review tools; earned **44 GitHub stars**

Git-Hired: AI Powered Adaptive Interview Platform

Nov 2025

- Shipped a **4-agent AI interview platform** (**FastAPI**, **React/TypeScript**, **WebSockets**) where a **5-tier scoring loop** adapts question difficulty ± 2 levels mid-interview and a **syntax-agnostic evaluator** scores code logic; auto-delivered 100-point reports to hiring managers over **SMTP**